A CASE-STUDY BASED REPORT

ON

Customer Purchase Prediction Using Logistic Regression and Random Forest Algorithms

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

Department of Computer Science and Information Technology

A.Y.: 2026-27

by

Name : B. Sathish Kumar	**roll :** 2520090008
Name : A. Shashank Reddy	**roll :** 2520090139
Name : P. Rahul	**roll :** 2520090016

Under the Esteemed guidance of

Dr. Y. Lakshmi Prasanna

Assistant Professor, Department of CS&IT

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

INDEX

Abstract **3**

1. Introduction **4**

1.1 Background 4

1.2 Motivation for the Project 4

1.3 Overview of the Work Carried Out 4

1.4 Scope and Responsible-Use Statement 5

2. Problem Statement **6**

2.1 Case-Study Scenario 6

2.2 Central Case-Study Question 6

2.3 Formal Problem Definition 6

2.4 Objectives that Guide the Study 7

2.5 Assumptions and Limitations 7

3. Dataset and Exploratory Data Analysis **8**

3.1 About the Dataset 8

3.2 Purpose of EDA 9

3.3 Preparation Steps Actually Performed 9

3.4 EDA Exploration – Implementation 10

3.5 Summary of EDA Findings 14

4. Algorithm Selection and Implementation **16**

4.1 Logistic Regression 16

4.2 Random Forest Classifier 17

4.3 Data Preprocessing 18

4.4 Training and Testing 19

4.5 Model Evaluation Metrics 20

4.6 Comparison of the Algorithms 21

4.7 Responsible Use and Limitations 24

5. Conclusion **26**

References **27**

Signature of the Team Members **27**

Abstract

Customer purchase prediction is a supervised machine learning task in which historical customer records are used to estimate whether a customer is likely to purchase a product. This case study applies Logistic Regression and Random Forest classifiers to a dataset of 1,000 customer records containing eight columns: age, gender, income, spending score, previous purchase, online activity, number of discounts and the purchase outcome. The dataset was inspected for missing values and duplicate records, and none were found. Categorical attributes were one-hot encoded, numerical attributes were standardised for Logistic Regression inside a pipeline fitted only on training data, and the records were divided into stratified training (80%) and testing (20%) sets.

Exploratory data analysis showed that 274 of the 1,000 customers (27.4%) made a purchase, so the target variable is moderately imbalanced. Purchase rates were higher among customers with high online activity and with a previous purchase, and both income and spending score were positively associated with purchase (correlation coefficients of 0.26 and 0.31) as well as with each other (0.50).

Both models were evaluated on the held-out test set using accuracy, precision, recall, F1-score and the confusion matrix, supplemented by five-fold cross-validation. Logistic Regression obtained an accuracy of 78.5% and an F1-score of 0.482, while Random Forest obtained 77.0% and 0.425. The two models differ by only a few test records, and both identify fewer than half of the actual purchasers at the default threshold of 0.5, so the findings are reported as indicative rather than conclusive. The study illustrates how preprocessing, exploratory analysis and metric-based evaluation together support data-driven customer analysis, and it identifies threshold adjustment, hyper-parameter tuning and richer behavioural features as directions for improvement.

Keywords: customer purchase prediction, supervised learning, Logistic Regression, Random Forest, exploratory data analysis, model evaluation.

1. Introduction

1.1 Background

Organisations that sell products to consumers routinely collect data about the people they serve, such as their age, income, browsing behaviour and earlier purchases. Machine learning offers a systematic way of turning such records into predictions. In a supervised learning problem, a model is trained on examples for which the correct outcome is already known and then used to predict the outcome for new examples. When the outcome belongs to one of a small number of categories, the problem is called classification.

Customer purchase prediction is a binary classification problem: for each customer, the model estimates whether the customer will purchase a product ("Yes") or will not ("No"). The estimate is based on demographic attributes, such as age and gender, and behavioural attributes, such as spending score, online activity and previous purchase history.

1.2 Motivation for the Project

Marketing effort, whether in the form of advertisements, discounts or personal communication, is limited by cost and by customer attention. If a business can estimate which customers are more likely to buy, it can direct its effort more deliberately instead of treating all customers identically. A purchase-prediction model is one tool for supporting that decision.

The project is also motivated by a methodological question that is central to machine learning practice: when should a simple, interpretable model be preferred over a more flexible one? Logistic Regression is simple and explains how each attribute influences the outcome. Random Forest is more flexible and can capture non-linear relationships and interactions between attributes, but is harder to interpret. Comparing the two on the same data, under the same evaluation procedure, gives a practical illustration of this trade-off.

1.3 Overview of the Work Carried Out

The work reported here followed the standard supervised-learning workflow. A customer dataset of 1,000 records was loaded and inspected; its structure, data types, missing values and duplicate records were examined; exploratory data analysis (EDA) was carried out using summary statistics and visualisations; and the data were split into training and testing subsets. A Logistic Regression model and a Random Forest model were trained on the training subset and evaluated on the unseen testing subset using accuracy, precision, recall, F1-score and the

confusion matrix. Five-fold cross-validation was used as an additional check on the stability of the results.

The implementation used Python 3.12 with the Pandas, NumPy, Matplotlib and Scikit-learn libraries (Pandas 3.0.2, NumPy 2.4.4, Matplotlib 3.10.8 and Scikit-learn 1.8.0) [1]. All dataset statistics, figures and evaluation results in this report were produced by running this code on the supplied CSV file.

1.4 Scope and Responsible-Use Statement

Scope. The study is restricted to a single tabular dataset and to two classification algorithms. It covers data inspection, preprocessing, EDA, model training, evaluation and comparison. It does not include hyper-parameter optimisation, deployment of a model into a production system, or causal analysis of why customers purchase.

Responsible use. A prediction produced by a model is a statistical estimate and not a certainty about an individual. The conclusions of this report apply only to the dataset analysed, and they should not be generalised to other customer populations without fresh validation. Customer data should be used lawfully and only for purposes consistent with the expectations of the customers concerned, and model outputs should support, rather than replace, informed human judgement. These points are discussed further in Section 4.7.

2. Problem Statement

2.1 Case-Study Scenario

Consider a retail business that maintains records of 1,000 customers. For each customer the business knows the age, gender, income and a spending score, whether the customer has bought from the business before, how actively the customer engages with the business online (low, medium or high), and the number of discounts associated with the customer's record. For every customer the business also knows whether a purchase of the product under consideration was made.

The business wishes to use these historical records to decide which kinds of customers are more likely to purchase, so that future promotional effort can be planned with better information. It has two candidate approaches in mind: an interpretable statistical model, Logistic Regression, and a more flexible ensemble model, Random Forest.

2.2 Central Case-Study Question

Given the demographic and behavioural attributes of a customer, how reliably can a supervised classification model predict whether the customer will purchase a product, and does Random Forest predict more reliably than Logistic Regression on this dataset?

2.3 Formal Problem Definition

Let the dataset contain n = 1,000 customer records. Each record consists of a vector of input features and a label. The feature vector of the i-th customer is denoted by x_i, and its label by y_i, where y = 1 indicates that the customer purchased and y = 0 indicates that the customer did not. In this study the feature vector contains seven attributes (Age, Gender, Income, Spending Score, Previous Purchase, Online Activity and Discounts), which become eight numerical inputs after categorical encoding. The goal is to learn a function f that maps a feature vector to a predicted label:

$$\hat{y} = f(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^d, \quad \hat{y} \in \{0, 1\} \tag{2.1}$$

Here x is the encoded feature vector, d is the number of encoded features (d = 8 in this study), $\hat{y}$ is the predicted label and $\mathbb{R}$ denotes the real numbers. The function f is learned from a training subset of the data and judged by how well its predictions agree with the known labels of a separate testing subset that was not used during training.

2.4 Objectives that Guide the Study

1. To inspect the customer dataset and verify its structure, data types, missing values and duplicate records.
2. To carry out exploratory data analysis and identify which attributes appear to be related to purchase behaviour.
3. To prepare the data for modelling by encoding categorical variables, scaling numerical variables where required, and splitting the data into training and testing sets without data leakage.
4. To train Logistic Regression and Random Forest classifiers on the training data.
5. To evaluate both models using accuracy, precision, recall, F1-score and the confusion matrix.
6. To compare the two algorithms on measured results and discuss which is more suitable for the problem, together with the limitations of the comparison.

2.5 Assumptions and Limitations

- The dataset is assumed to be a reasonable sample of the customers about whom predictions are required. Its collection method and original source are not documented in the material supplied, so this assumption cannot be verified.
- The label “Purchased” is assumed to be recorded accurately, and the meaning of each attribute is taken from its column name. In particular, the currency of Income and the exact meaning of Discounts (an integer between 0 and 5) are not documented.
- The records are assumed to be independent of one another, and the relationship between attributes and purchase is assumed to remain stable over time.
- The dataset is small (1,000 records), and the testing subset contains only 200 records, of which 55 are purchasers. Differences between models of a few records are therefore not strong evidence of a real difference.
- Default or lightly specified model settings were used and no systematic hyper-parameter tuning was performed. A single train–test split with a fixed random seed was used for the main comparison.
- The study identifies statistical association, not causation.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The dataset was supplied as a comma-separated values (CSV) file named *customer_purchase_prediction_1000_csv.csv*. The source of the data (for example, a public repository, a company, or a synthetic generator) is not stated in the material provided and has not been assumed. It contains **1,000 records and 8 columns** (seven input attributes and one target). Each row represents one customer. Table 3.1 describes the columns.

Table 3.1: Columns of the dataset

Column	Data type	Kind	Role	Values observed
Age	Integer	Numerical	Feature	18 to 60 years
Gender	Text	Categorical (nominal)	Feature	Male, Female
Income	Integer	Numerical	Feature	20,000 to 110,888
Spending Score	Integer	Numerical	Feature	1 to 100
Previous Purchase	Text	Categorical (binary)	Feature	Yes, No
Online Activity	Text	Categorical (ordinal)	Feature	Low, Medium, High
Discounts	Integer	Numerical (discrete)	Feature	0 to 5
Purchased	Text	Categorical (binary)	Target	Yes, No

There are four numerical attributes (Age, Income, Spending Score and Discounts) and four categorical attributes (Gender, Previous Purchase, Online Activity and Purchased). The target variable is **Purchased**. Every column was found to be complete: the count of missing values was zero in all eight columns. A check for fully duplicated rows found **no duplicate records**. Table 3.2 gives descriptive statistics for the numerical attributes.

Table 3.2: Descriptive statistics of the numerical attributes (n = 1,000)

Attribute	Mean	Std. deviation	Minimum	Median	Maximum
Age	39.01	12.33	18	40	60
Income	55,620.14	17,520.87	20,000	55,702.5	110,888

Attribute	Mean	Std. deviation	Minimum	Median	Maximum
Spending Score	50.92	18.77	1	51	100
Discounts	2.46	1.71	0	2	5

Among the 1,000 customers, 518 are male and 482 are female; 510 have made a previous purchase and 490 have not; and online activity is Medium for 444 customers, Low for 289 and High for 267. The target is imbalanced to a moderate degree: 274 customers (27.4%) purchased and 726 (72.6%) did not. A model that always predicted "No" would therefore achieve an accuracy of 72.6% on this dataset, which is an important reference point when judging the accuracy figures reported later.

3.2 Purpose of EDA

Exploratory data analysis is the examination of a dataset, using summary statistics and graphs, before models are built. Its purposes in this study are: (i) to verify data quality, by checking types, missing values, duplicates and plausible ranges; (ii) to understand the distribution of each attribute and of the target; (iii) to detect class imbalance, which affects how models should be evaluated; (iv) to explore relationships between the attributes and the target, and between the attributes themselves; and (v) to inform preprocessing decisions, such as which attributes need encoding or scaling.

Several standard graphs are used. A **bar chart** shows counts or rates for categories and is suitable for the target variable and for categorical attributes. A **histogram** divides a numerical attribute into intervals and shows how its values are distributed, revealing skewness and unusual values. A **box plot** summarises the median, quartiles and spread of a numerical attribute for each target class, allowing the two classes to be compared. A **correlation heatmap** displays the pairwise correlation coefficients between numerical variables as a colour-coded matrix; coefficients close to +1 or −1 indicate strong linear association and values near 0 indicate little linear association. Correlation measures linear association only and does not imply causation.

3.3 Preparation Steps Actually Performed

The following steps were carried out on the supplied dataset.

1. The CSV file was loaded into a Pandas DataFrame and its shape (1,000 rows, 8 columns) and data types were checked.

2. Missing values were counted column by column. None were found, so no imputation or row deletion was necessary.
3. Duplicate rows were counted. None were found, so no records were removed.
4. Categorical value counts were examined to confirm that the categories were consistent (for example, no spelling variants).
5. The target column Purchased was converted to a numerical label (Yes = 1, No = 0). The remaining seven columns formed the feature set.
6. The data were divided into a training set of 800 records and a testing set of 200 records using a stratified split (random seed 42), so that both subsets keep the original proportion of purchasers (219 of 800 in training, 55 of 200 in testing).
7. Categorical features were one-hot encoded and, for Logistic Regression only, numerical features were standardised. Both operations were placed inside a Scikit-learn pipeline so that they were learned from training data alone (Section 4.3).

3.4 EDA Exploration – Implementation

The EDA was implemented in Python. The core commands used for inspection and for the graphs are shown below in condensed form.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("customer_purchase_prediction_1000_csv.csv")
print(df.shape, df.dtypes, df.isna().sum(), df.duplicated().sum())
print(df.describe())
df["y"] = (df["Purchased"] == "Yes").astype(int)
print(df["Purchased"].value_counts())
print(df.groupby("Online Activity")["y"].mean())
print(df[["Age","Income","Spending Score","Discounts","y"]].corr())
df["Income"].hist(bins=20)      # histogram
```

3.4.1 Distribution of the target variable

Figure 3.1 shows the number of purchasers and non-purchasers. The “No” class (726 customers) is about 2.6 times as large as the “Yes” class (274 customers). This imbalance means that accuracy alone can be misleading, and precision, recall and F1-score for the “Yes” class must also be examined.

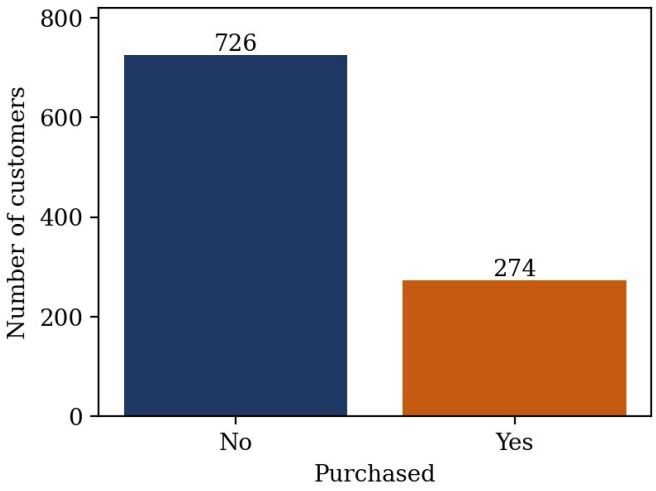


Figure 3.1: Distribution of the target variable (Purchased)

3.4.2 Distribution of numerical attributes

Figure 3.2 presents histograms of Age, Income and Spending Score. Ages range from 18 to 60 and are spread across the whole range rather than concentrated around a single value. Income and Spending Score are centred near their means (about 55,620 and 51 respectively); Income has a right tail extending to a maximum of 110,888 and a lower bound of 20,000.

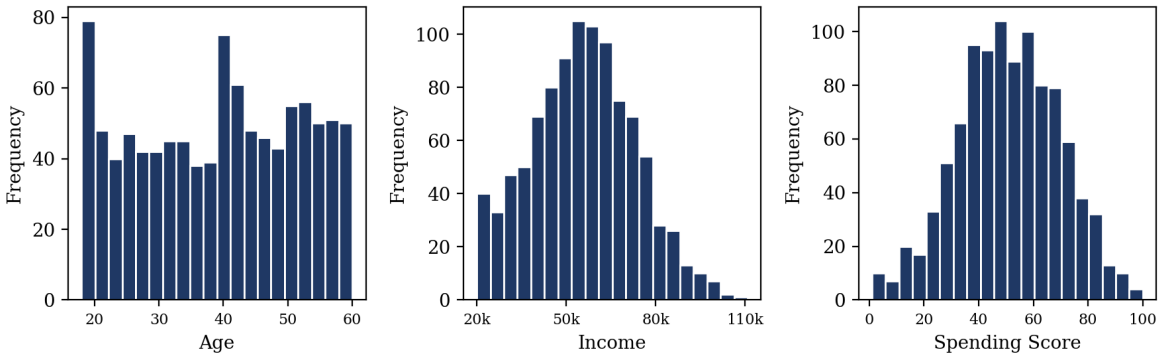


Figure 3.2: Histograms of Age, Income and Spending Score

3.4.3 Purchase behaviour across categorical attributes

Figure 3.3 and Table 3.3 show the percentage of customers who purchased within each category of Gender, Previous Purchase, Online Activity and Discounts.



Figure 3.3: Purchase rate (%) by categorical attribute

Table 3.3: Purchase rate within each category

Attribute	Category	Customers	Purchase rate
Gender	Female	482	26.1%
	Male	518	28.6%
Previous Purchase	No	490	20.6%
	Yes	510	33.9%
Online Activity	Low	289	18.0%
	Medium	444	26.8%
	High	267	38.6%
Discounts	0	162	27.2%
	1	188	27.1%
	2	168	23.2%
	3	157	28.0%
	4	156	32.7%
	5	169	26.6%

The purchase rate rises steadily from Low (18.0%) to Medium (26.8%) to High (38.6%) online activity, and is higher for customers with a previous purchase (33.9%) than for those without (20.6%). The difference between female (26.1%) and male (28.6%) customers is small. The purchase rate does not change in a consistent direction with the number of discounts (it ranges between 23.2% and 32.7% across the six values), so no clear relationship between Discounts and purchase is visible in this simple view.

3.4.4 Numerical attributes by purchase outcome

Figure 3.4 compares the distribution of Age, Income and Spending Score for non-purchasers and purchasers, and Table 3.4 gives the corresponding averages.

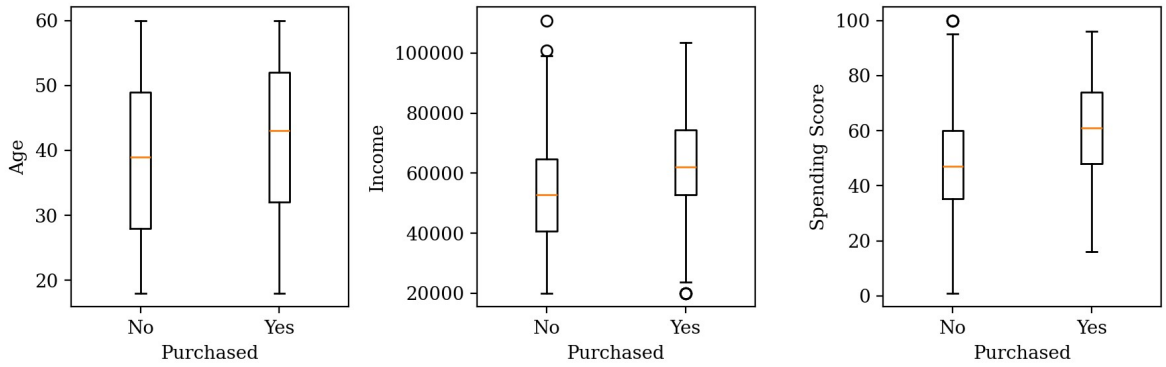


Figure 3.4: Box plots of Age, Income and Spending Score by purchase outcome

Table 3.4: Mean and median of numerical attributes by purchase outcome

Attribute	Mean (No)	Mean (Yes)	Median (No)	Median (Yes)
Age	38.23	41.10	39	43
Income	52,815	63,052	52,851	62,121
Spending Score	47.32	60.46	47	61

Customers who purchased have a higher average income (63,052 against 52,815) and a higher average spending score (60.5 against 47.3) than those who did not, and are on average about three years older (41.1 against 38.2). The boxes in Figure 3.4 overlap considerably, which indicates that none of these attributes separates the two classes cleanly on its own.

3.4.5 Correlation analysis

Figure 3.5 shows the correlation heatmap for the numerical attributes and the numerically encoded target.

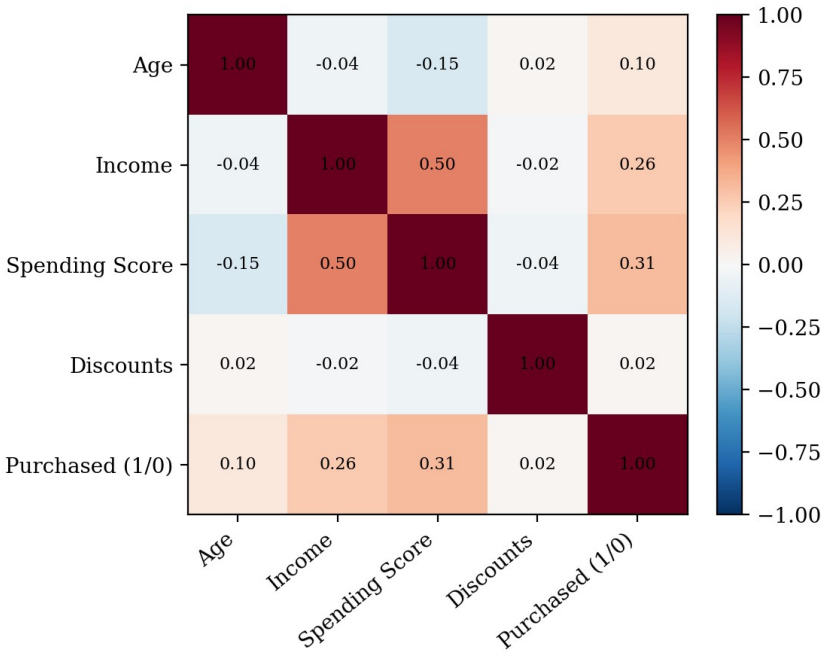


Figure 3.5: Correlation heatmap of numerical attributes and the purchase label

Income and Spending Score are moderately positively correlated (0.50), so the two attributes carry partly overlapping information. Spending Score (0.31) and Income (0.26) have the strongest correlations with the purchase label, followed by Age (0.10); the correlation of Discounts with purchase is close to zero (0.02). Age is weakly negatively correlated with Spending Score (−0.15). All of the correlations with the target are modest, which suggests that purchase is not determined by any single numerical attribute and that a good model can be expected to achieve only moderate predictive performance.

3.5 Summary of EDA Findings

- The dataset has 1,000 records and 8 columns, with no missing values and no duplicate rows.
- The target is moderately imbalanced: 27.4% purchasers and 72.6% non-purchasers.
- Higher online activity and a previous purchase are associated with higher purchase rates.
- Purchasers tend to have higher income and higher spending score, and are slightly older on average.
- Income and Spending Score are moderately correlated with each other (0.50), so multicollinearity should be kept in mind when interpreting Logistic Regression coefficients.

- Gender and Discounts show little or no consistent relationship with purchase in the descriptive analysis.
- No individual attribute separates purchasers from non-purchasers cleanly; only moderate predictive performance should be expected.

4. Algorithm Selection and Implementation

Two classifiers were selected. Logistic Regression is a standard baseline for binary classification: it is fast, has few settings, and produces coefficients that can be interpreted. Random Forest is an ensemble method that handles non-linear relationships and interactions without manual feature engineering. Using both allows a direct comparison between an interpretable linear model and a more flexible non-linear model on identical data.

4.1 Logistic Regression

Binary classification. Logistic Regression is a model for predicting a binary outcome. Despite the word "regression" in its name, it is used for classification: it estimates the probability that a record belongs to the positive class (here, "customer purchases").

Sigmoid function and probability estimation. The model first forms a weighted sum of the input features, and then converts this sum into a probability using the sigmoid (logistic) function:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_d x_d \tag{4.1}$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{4.2}$$

$$P(y = 1 \mid \mathbf{x}) = \sigma(z) \tag{4.3}$$

Here $x_1, \ldots, x_d$ are the feature values, $\beta_1, \ldots, \beta_d$ are the coefficients (weights) learned from the training data, β_0 is the intercept, z is the linear score, e is Euler's number and $P(y = 1 \mid x)$ is the probability that the customer purchases, given the features x. The sigmoid maps any real number z to a value between 0 and 1, which makes the output interpretable as a probability: large positive z gives a probability near 1, large negative z gives a probability near 0, and z = 0 gives 0.5. The coefficients are found by minimising the log-loss (cross-entropy) on the training data, which penalises confident wrong predictions heavily:

$$L = -\frac{1}{n} \sum_{i=1}^{n} [\, y_i \ln p_i + (1 - y_i) \ln(1 - p_i) \,] \tag{4.4}$$

In Equation 4.4, n is the number of training records, y_i is the true label of record i and p_i is its predicted probability of purchase.

Classification threshold. A probability is converted into a class label by comparing it with a threshold t. The conventional value is t = 0.5: the model predicts "Yes" if $P(y = 1 \mid x) \geq 0.5$ and "No" otherwise. The threshold is a design choice. Lowering it makes the model predict "Yes" more often, which raises recall at the cost of precision; raising it has the opposite effect. This is particularly relevant for imbalanced data.

Advantages. The model is simple, fast to train and gives well-defined probabilities. Each coefficient describes the direction and strength of a feature's influence on the log-odds of purchase, which makes the model easy to explain. It also tends to be stable on small datasets.

Limitations. The model assumes that the log-odds of purchase are a linear combination of the features, so it cannot capture non-linear effects or interactions unless they are added as features explicitly. Strongly correlated features, such as Income and Spending Score here, can make individual coefficients unstable. Numerical features should be scaled for the optimisation to behave well and for coefficients to be comparable.

4.2 Random Forest Classifier

Ensemble learning. An ensemble combines the predictions of several models to obtain a more reliable prediction than a single model would give. Random Forest [2] is an ensemble of decision trees. A decision tree classifies a record by asking a sequence of questions about its features (for example, "Is Spending Score above 55?"), each of which splits the data into groups, until a final prediction is reached at a leaf. A single deep tree can fit the training data very closely but is sensitive to small changes in that data, a problem known as overfitting.

Bootstrap sampling. To build a forest of T trees, each tree is trained on a bootstrap sample: a sample of n records drawn from the training data at random with replacement. Each tree therefore sees a slightly different version of the data, and the trees differ from one another.

Feature selection at splits. At each split in each tree, only a random subset of the features is considered as candidates (for classification, commonly the square root of the number of features). The best split among these candidates is chosen according to an impurity measure such as the Gini impurity:

$$G = 1 - \sum_{k} p_k^2 \tag{4.5}$$

where p_k is the proportion of records of class k in a node; G is 0 for a perfectly pure node and larger for mixed nodes. Restricting each split to a random subset of features makes the trees less alike, and so reduces the correlation between their errors.

Aggregation of predictions. For a new record, every tree makes a prediction, and the forest returns the class chosen by the majority of trees. In the Scikit-learn implementation, the forest averages the class probabilities estimated by the individual trees:

$$P(y=1 \mid \mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} p_t(\mathbf{x}) \tag{4.6}$$

Here T is the number of trees and $p_t(x)$ is the probability of purchase estimated by tree t for features x. Averaging many different trees reduces variance and usually limits overfitting compared with a single tree.

Advantages. Random Forest can model non-linear relationships and interactions, needs no feature scaling, is relatively robust to outliers, and provides feature-importance scores.

Limitations. The model is less interpretable than Logistic Regression, requires more memory and training time, and can still overfit, particularly when trees are grown to full depth on a small dataset. Impurity-based feature importances tend to favour continuous attributes with many possible split points over binary attributes.

4.3 Data Preprocessing

Handling missing values. Many algorithms cannot accept missing values. Common remedies are to remove affected records or to impute values, for example with the median for numerical features and the most frequent category for categorical ones. In this dataset no values were missing, so neither was required. In a pipeline intended for new data, an imputation step would be added so that incomplete records can still be processed, with imputation statistics learned from the training set only.

Encoding categorical variables. Machine learning algorithms operate on numbers, so text categories must be converted. One-hot encoding replaces a categorical column with one binary column per category. To avoid redundancy, one category is dropped as the reference ("drop first"). Gender became *Gender_Male* (reference: Female), Previous Purchase became *Previous Purchase_Yes* (reference: No), and Online Activity became *Online Activity_Low* and *Online Activity_Medium* (reference: High). The target was mapped to Yes = 1 and No = 0. Together with the four numerical attributes, this gives d = 8 model inputs.

Scaling. Logistic Regression is sensitive to the scale of its inputs: Income is measured in tens of thousands while Discounts lies between 0 and 5. The four numerical attributes were therefore standardised (each value replaced by its distance from the mean, in units of the standard deviation), which also lets the coefficients be compared in size. Random Forest splits on thresholds and is unaffected by monotonic scaling, so its features were left unscaled.

Avoiding data leakage. Data leakage occurs when information from the testing data influences model training, which makes the measured performance unrealistically optimistic. To prevent this, the train–test split was performed first. The scaler and the encoder were then fitted on the training data only and merely applied to the testing data, and they were wrapped together with the classifier in a single pipeline. The same pipeline structure was used in cross-validation, so that preprocessing was re-fitted within each training fold.

4.4 Training and Testing

The dataset was split into 80% training data (800 records) and 20% testing data (200 records). The split was stratified on the target so that the proportion of purchasers is the same in both subsets: 219 of 800 (27.4%) in training and 55 of 200 (27.5%) in testing. A fixed random seed (42) was used so that the results can be reproduced. The model settings are given in Table 4.1.

Table 4.1: Model settings used (Scikit-learn)

Item	Logistic Regression	Random Forest
Preprocessing	One-hot encoding (drop first) + standardisation of numerical features	One-hot encoding (drop first); no scaling
Main settings	max_iter = 1000; default L2 regularisation (C = 1.0)	n_estimators = 200; other settings at default values
Decision threshold	0.5	0.5 (majority/average probability)
Random seed	42	42

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

X = df.drop(columns=["Purchased", "y"]);  y = df["y"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42)

num = ["Age", "Income", "Spending Score", "Discounts"]
```

```
cat = ["Gender", "Previous Purchase", "Online Activity"]
prep_lr = ColumnTransformer([("n", StandardScaler(), num),
                             ("c", OneHotEncoder(drop="first"), cat)])
prep_rf = ColumnTransformer([("n", "passthrough", num),
                             ("c", OneHotEncoder(drop="first"), cat)])
lr = Pipeline([("p", prep_lr),
               ("m", LogisticRegression(max_iter=1000, random_state=42))])
rf = Pipeline([("p", prep_rf),
               ("m", RandomForestClassifier(n_estimators=200, random_state=42))])
lr.fit(X_train, y_train);  rf.fit(X_train, y_train)
```

4.5 Model Evaluation Metrics

Evaluation uses the confusion matrix, which counts the model's predictions on the testing data against the true labels. Taking "purchase" (Yes) as the positive class, its four cells are defined as follows.

- **True Positive (TP):** the customer purchased and the model predicted a purchase.
- **True Negative (TN):** the customer did not purchase and the model predicted no purchase.
- **False Positive (FP):** the customer did not purchase but the model predicted a purchase (a false alarm).
- **False Negative (FN):** the customer purchased but the model predicted no purchase (a missed buyer).

Table 4.2: Structure of the confusion matrix

	Predicted: Yes	**Predicted: No**
Actual: Yes	True Positive (TP)	False Negative (FN)
Actual: No	False Positive (FP)	True Negative (TN)

The following metrics are computed from these four counts.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.7}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.8}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4.9}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.10}$$

Accuracy is the fraction of all predictions that are correct. It is easy to understand but can be misleading when classes are imbalanced, since a model that always predicts the majority class can obtain a high accuracy. **Precision** is the fraction of predicted purchasers who actually purchased; it matters when contacting a customer who will not buy is costly. **Recall** is the fraction of actual purchasers whom the model identifies; it matters when missing a potential buyer is costly. The **F1-score** is the harmonic mean of precision and recall and is high only when both are reasonably high, so it summarises performance on the positive class in a single value. As an additional threshold-independent measure, the area under the ROC curve (ROC-AUC) is also reported; a value of 0.5 corresponds to random ranking and 1.0 to perfect ranking.

4.6 Comparison of the Algorithms

Both models were trained on the 800 training records and evaluated on the 200 testing records. The testing set contains 145 non-purchasers and 55 purchasers. Table 4.3 presents the measured results, and Figure 4.1 shows the confusion matrices.

Table 4.3: Test-set performance of the two models (threshold = 0.5; positive class = Yes)

Metric	Logistic Regression	Random Forest
Accuracy	0.785 (78.5%)	0.770 (77.0%)
Precision	0.714	0.680
Recall	0.364	0.309
F1-score	0.482	0.425
ROC-AUC	0.760	0.718
True Positives (TP)	20	17
True Negatives (TN)	137	137
False Positives (FP)	8	8
False Negatives (FN)	35	38

Reference point: a classifier that always predicts “No” would obtain an accuracy of 72.5% on this testing set (145 of 200).

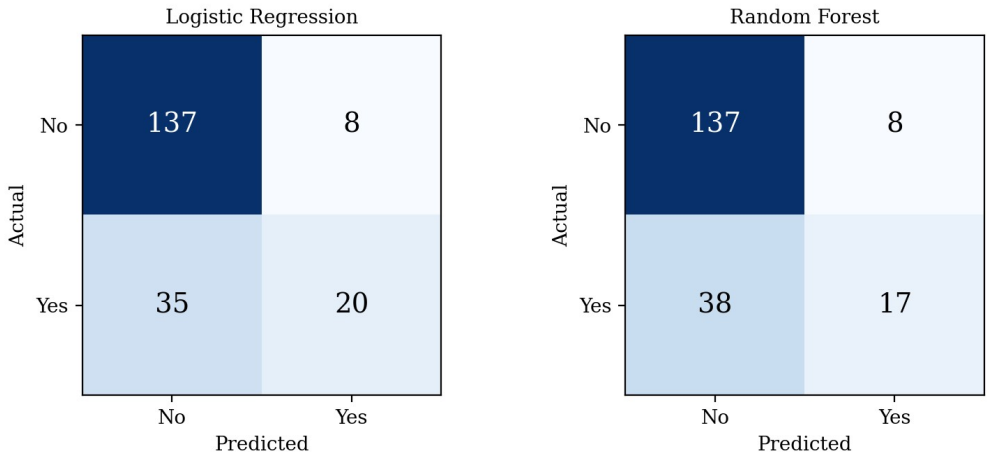


Figure 4.1: Confusion matrices of Logistic Regression and Random Forest on the testing set

To check that the result did not depend on one particular split, five-fold stratified cross-validation was performed on the whole dataset, and the accuracy of each model on its own training data was also recorded (Table 4.4).

Table 4.4: Cross-validation results and training accuracy

Measure	Logistic Regression	Random Forest
5-fold CV mean accuracy	0.756	0.736
5-fold CV mean F1-score	0.411	0.347
Accuracy on training set	0.745	1.000
Accuracy on testing set	0.785	0.770

4.6.1 Interpretation of the model internals

The Logistic Regression coefficients (Table 4.5) are on the log-odds scale; numerical features were standardised, so a coefficient is the change in log-odds for a one-standard-deviation increase in that feature, holding the others fixed. Spending Score (0.74) and Previous Purchase = Yes (0.80) have the strongest positive influence among the features other than Online Activity, whose Low (−1.10) and Medium (−0.46) levels reduce the log-odds of purchase relative to High. Age (0.36) and Income (0.28) have smaller positive coefficients. The coefficients for Discounts (0.20) and Gender_Male (0.20) are the smallest, which is consistent with the weak relationships seen in the EDA. Because Income and Spending Score are correlated, their individual coefficients should be interpreted with caution.

Table 4.5: Logistic Regression coefficients and Random Forest feature importances

Encoded feature	LR coefficient	RF importance
Spending Score	0.740	0.270
Income	0.278	0.266
Age	0.364	0.222
Discounts	0.203	0.105
Previous Purchase_Yes	0.796	0.044
Gender_Male	0.201	0.033
Online Activity_Medium	−0.463	0.030
Online Activity_Low	−1.097	0.030

The Random Forest feature importances (Figure 4.2) rank Spending Score, Income and Age highest. These scores are not directly comparable with the Logistic Regression coefficients: impurity-based importance favours continuous features with many possible split points, which probably explains the low importance given to the binary and three-level attributes even though the EDA showed that Online Activity and Previous Purchase are clearly associated with purchase.

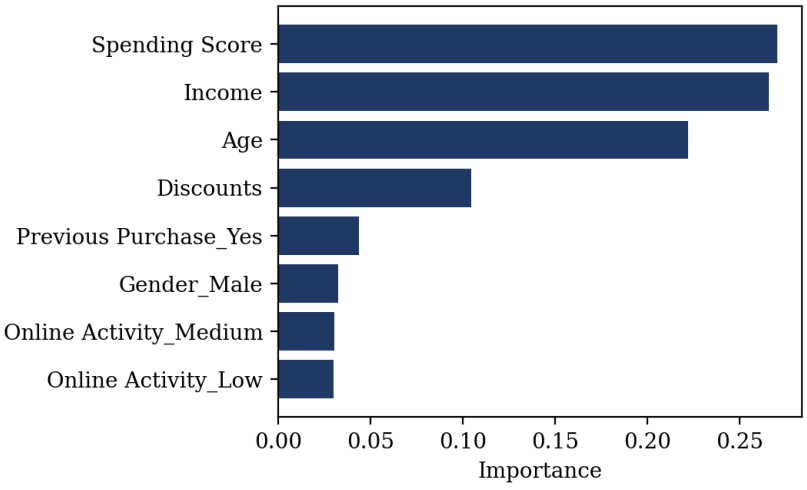


Figure 4.2: Random Forest feature importances

4.6.2 Discussion and model selection

On the testing set, Logistic Regression achieved higher accuracy (78.5% against 77.0%), precision, recall, F1-score and ROC-AUC than Random Forest, and it also scored higher in

cross-validation. The Random Forest classified every training record correctly (training accuracy 1.000) but performed noticeably worse on unseen records, which is a sign of overfitting. These measurements favour Logistic Regression on this dataset.

The result should nevertheless be interpreted with care. The two models differ by three records out of 200 in test accuracy, and both are only modestly above the 72.5% accuracy of the always-“No” reference. Both models also have low recall: at the 0.5 threshold, Logistic Regression identifies 20 of the 55 actual purchasers and Random Forest 17. This is consistent with the EDA, which found only moderate associations between the attributes and the target. No hyper-parameter tuning or threshold adjustment was performed, and a tuned Random Forest might perform differently. The measured evidence therefore supports a modest preference for Logistic Regression in this study, not a general statement that it is the better algorithm.

The final model should be chosen according to measured performance on data not used for training, together with the requirements of the application. If the aim is to explain which customer attributes influence purchase, the interpretable Logistic Regression is preferable. If the cost of missing a potential buyer is high, recall for the “Yes” class and ROC-AUC should carry more weight than accuracy, and the decision threshold should be lowered and the resulting precision–recall trade-off re-measured. If the cost of contacting non-buyers is high, precision should be given more weight. The selected model should finally be validated on new data before it is relied upon.

4.7 Responsible Use and Limitations

- **Predictions are estimates, not guarantees.** A model outputs a probability or a class based on patterns in past data. Even the better model in this study missed 35 of 55 purchasers in the test set, and it cannot determine what an individual customer will actually do.
- **Performance depends on data quality and representativeness.** The results apply to the dataset analysed. If the customers of interest differ from those in the data, or their behaviour changes over time, performance may decline, and the model should be re-evaluated and retrained.
- **Imbalanced data can affect evaluation.** With 27.4% purchasers, accuracy overstates the usefulness of a model, as the always-“No” reference shows. Precision, recall, F1-score and the confusion matrix should be read together, and class weighting, resampling or threshold tuning can be considered.

- **Biased historical data can produce biased predictions.** If past data reflect unequal treatment or unequal access to products, a model trained on it can reproduce that bias. Attributes such as age and gender in particular should be examined for fairness before any predictions are used to treat groups of customers differently.
- **Customer data should be handled responsibly.** Personal and financial information should be collected and used lawfully, with appropriate consent, restricted access and protection against misuse, and only for purposes that customers would reasonably expect.
- **Predictions should support, not replace, human decisions.** Model outputs should be treated as one input to a decision made by informed people who can consider context that the data do not contain and who remain accountable for the outcome.
- **Statistical limitations of this study.** A small dataset, a single train–test split and the absence of hyper-parameter tuning mean that the performance figures carry uncertainty and should not be over-interpreted.

5. Conclusion

This case study addressed the problem of predicting whether a customer will purchase a product from demographic and behavioural attributes. Such a prediction can help a business understand which customers are more likely to buy and plan its marketing effort with better information, although it can only provide estimates and not certainties.

Logistic Regression and Random Forest classifiers were implemented and compared on a dataset of 1,000 customer records. Preprocessing and EDA formed an essential first stage: the data were verified to have no missing values or duplicates, categorical variables were encoded, and numerical variables were scaled where needed, with all preprocessing fitted on training data only to avoid leakage. EDA revealed that 27.4% of customers purchased, that high online activity, a previous purchase, higher income and higher spending score were associated with purchase, and that no single attribute separated the classes cleanly.

Evaluation on a held-out testing set using accuracy, precision, recall, F1-score and the confusion matrix showed that Logistic Regression (accuracy 78.5%, F1-score 0.482) performed slightly better than Random Forest (accuracy 77.0%, F1-score 0.425), and the cross-validation results pointed in the same direction. However, the margin is small, both models are only modestly better than always predicting “No” in terms of accuracy, and both detect fewer than half of the actual purchasers at the default threshold. The study therefore shows the importance of looking beyond accuracy and of evaluating models on unseen data, rather than demonstrating a highly accurate predictor.

Several improvements to the predictive methodology are possible: tuning the hyper-parameters of both models (for example, tree depth and number of trees for Random Forest, and regularisation strength for Logistic Regression) with cross-validation; adjusting the decision threshold, or using class weights or resampling, to improve recall for purchasers; adding interaction or derived features; comparing further algorithms such as gradient boosting; examining model fairness across customer groups; and validating the chosen model on a larger and more recent dataset with documented provenance.

References

[1] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[2] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

Signature of the Team Members

B. Sathish Kumar	**A. Shashank Reddy**	**P. Rahul**
2520090008	2520090139	2520090016